

Hanning Lu

lhnjames.github.io | Google Scholar

Efficient and reliable LLM systems, agent reinforcement learning, and inference optimization.

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Leeds, United Kingdom

EDUCATION

University of Leeds

B.S. in Computer Science; expected first-class honours (1:1).

Leeds, United Kingdom

Sep 2024 – Jun 2027 (Expected)

RESEARCH EXPERIENCE

Compiler Optimization and LLVM IR Decompilation

Research Assistant | Supervised by Chunwei Xia, Zheng Wang

University of Leeds, UK

May 2025 – Sep 2026

- Developed agent-based C/C++ performance optimization; published in the **ACM ICS 2026 Workshop**.
- Developed **AICORE**, using LLVM remarks, IR probes, and runtime feedback to jointly optimize source rewrites, loop pragmas, and compiler parameters through a compile-validate-measure loop.
- Achieved **4.487×** / **1.299×** arithmetic-mean speedups over -O3 on 30 PolyBench / 29 cBench programs; all reported candidates passed output checks (in submission).
- Applied **iterative LLM self-correction** to LLVM IR decompilation, raising accuracy from ~50% to **over 90%** on obfuscated IR tests.
- Evaluated prompt framing in LLM cascades conditioned on ground truth, showing that more frequent revision does not imply more reliable revision (co-first-author preprint).

Agent Self-Improvement and Reinforcement Learning

Research Assistant | Supervised by Bin Chong

Peking University, China

Oct 2025 – Jul 2026

- Proposed **LJ-GQA**, a **gradient-guided agent self-improvement method** that reweights advantages using output-gradient agreement across semantically similar decisions from independent rollouts.
- Achieved **94.6% success** on ALFWorld with Qwen2.5-1.5B; trajectory audits identified exploration-loop deadlocks as the remaining failure mode (in submission).
- Combined a **verifiable evidence graph** with Qwen3-0.6B LoRA-SFT and Active-GRPO for social bot detection: 98.9% / 98.8% accuracy on Fox8-23 / BotSim-24 and near-100% evidence coverage.

Efficient LLM Inference and Model Optimization

Research Assistant | Supervised by Tong Yang

Peking University, China

Jun 2024 – Jun 2025

- Developed **adaptive token retention** across attention heads during generation, reducing KV-cache size by **70%** with ~1% performance loss on Llama-2-7B/LongBench.
- Studied quantization, structured/unstructured pruning, and sparse-weight inference for memory-efficient, low-latency deployment; co-authored a [quantization survey](#).

TPS-Bench: Multi-Tool Agent Planning and Scheduling

Research Assistant | Supervised by Zhijie Deng

Shanghai Jiao Tong Univ.

Jun 2025 – Aug 2025

- Contributed to **TPS-Bench**, an MCP-based benchmark and agent framework for task decomposition, tool selection, and coordinated multi-step execution.
- Used **adaptive tool reselection**, execution scheduling, and intermediate-result aggregation to improve multi-tool planning reliability and fault recovery. [Code](#).

Loop Transformers and Recurrent-Compute Reasoning

Research Assistant | Supervised by Menglin Yang

HKUST (Guangzhou)

May 2026 – Sep 2026

- Compared Frozen Loop, equal-magnitude noise, geometric steering, LoRA, and LoRA Loop on WinoGrande, bAbI, and ProofWriter; frozen-layer repetition did not consistently outperform noise.
- Used these controlled comparisons to identify **learned task-aligned recurrent updates** as the effective mechanism, rather than simply repeating frozen layers.

Knowledge Graph Reasoning and Voice AI

Research Assistant | Supervised by Jun Wei

Tsinghua University, China

Jun 2025 – Aug 2025

- Combined **structured graph traversal** with cross-node evidence aggregation to support multi-hop knowledge-graph reasoning for complex question answering.
- Integrated LLM inference, incremental generation, streaming TTS, and audio delivery into a serving pipeline for **low-latency, real-time voice interaction**.

INTERNSHIP EXPERIENCE

Xingjie Intelligence

Video Generation Infrastructure Engineer Intern

AI Startup

Jun 2026 – Sep 2026

- Built **SGLang-based serving** for Wan diffusion models using tensor, pipeline, and sequence parallelism, KV caching, and block-wise autoregressive generation.
- Profiled GPU bottlenecks with Nsight Systems, Nsight Compute, and PyTorch Profiler to guide optimization.
- Evaluated FP8, weight-only/activation quantization, CUDA Graphs, kernel fusion, continuous batching, and streaming inference using FPS, chunk latency, TTFT, memory use, and scaling efficiency.

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Artificial Intelligence Engineer Intern

AI Startup

Jun 2025 – Sep 2025

- Designed **multi-agent coordination** and configurable task-execution workflows to improve data-collection accuracy, runtime stability, and fault recovery.

SELECTED PUBLICATIONS

1. **H. Lu**, C. Xia, Z. Wang. AICORE: Compiler-Guided Joint Source and Parameter Optimization. (in submission).
2. **H. Lu**, C. Xia. Agent-Agnostic End-to-End C/C++ Application Performance Optimization. Proceedings of the 40th ACM International Conference on Supercomputing Workshops, 2026. DOI: [10.1145/3774895.3812199](https://doi.org/10.1145/3774895.3812199).
3. T. Liang, Y. Liu, S. Luo, Y. Yang, **H. Lu**, Y. Huang, J. Zhang, Y. Qi, K. Xu, C. Ran, B. Pan, B. Chong. Granularity-Adaptive Credit Assignment for Long-Horizon LLM Agent Reinforcement Learning. (in submission).
4. K. Hu, Y. Yang, **H. Lu**, Y. Chen, C. Zhang, H. Zheng, K. Xu, C. Ran, H. Peng, B. Chong. Action, Not Belief: What Makes Hypothesis Prompting Work in Debugger Agents. (in submission).
5. W. Zhong*, **H. Lu***, Z. Yao, S. Zhang, C. Xia. Revision Propensity Is Not Revision Reliability: Ground-Truth-Conditioned Evaluation of Prompt Framing in LLM Cascades. Preprint, 2026.
6. Y. Wang, T. Yang, X. Liang, G. Wang, **H. Lu**, Z. Xu, Y. Li, W. Li. Art and Science of Quantizing Large-Scale Models: A Comprehensive Overview. arXiv preprint, 2024. [arXiv:2409.11650](https://arxiv.org/abs/2409.11650).
7. H. Jia, Y. Liu, B. Chong, Y. Yang, Y. Chen, J. Liang, Q. Li, **H. Lu**, K. Xu, H. Zheng, C. Zhang, H. Peng, P. S. Yu. FinHarness: An Inline Lifecycle Safety Harness for Finance LLM Agents. arXiv preprint, 2026. [arXiv:2605.27333](https://arxiv.org/abs/2605.27333). (in submission).
8. F. Zou, S. Luo, Y. Yang, Y. Liu, Y. Bai, **H. Lu**, C. Zhang, H. Zheng, K. Xu, C. Ran, B. Chong. When Correct Predictions Require Incorrect States: A Compatibility Audit of Physical State Interfaces. (in submission).
9. **H. Lu**, Y. Yang, J. Su, Y. Liu, Z. Yao, Y. Li, T. Liang, Z. Zhang, R. Ran, K. Xu, B. Chong. SAHG: Sector-Anisotropic Hyperbolic Graph Model for Social Bot Detection. arXiv preprint, 2026. [arXiv:2605.30166](https://arxiv.org/abs/2605.30166). (in submission).
10. **H. Lu**, K. Hu, Y. Yang, Y. Liu, Z. Yao, Y. Li, Y. Chen, C. Zhang, H. Zheng, K. Xu, C. Ran, B. Chong. VeriBot: Evidence-Routed Social Bot Detection with Auditable Decision Chains. (In submission).
11. K. Hu, Y. Yang, **H. Lu**, Y. Liu, Y. Chen, C. Zhang, H. Zheng, K. Xu, C. Ran, H. Peng, P. S. Yu, B. Chong. BotRoute: Dual-Branch Selective Routing for Social Bot Detection. (In submission).
12. Z. Yao, Y. Tang, **H. Lu**, Y. Li, M. Yang. Hierarchy-Aware Sparse Autoencoders via Activation-Cone Routing. (In submission).
13. Z. Yao, **H. Lu**. HCGM: Hyperbolic Code Graph Modeling for Repository-Level Software Intelligence. (In submission).

* Co-first authors.

COMPETITIONS & HONORS

Gold Medal, CCPC Zhengzhou Invitational – Team Captain

2024

Silver Medal, 48th ICPC Xi'an Invitational – Team Captain